

Agricultural Index Insurance: An Optimization Approach

José I. Velarde Morales Linwei Xin

University of Chicago
Booth School of Business

July 21, 2025

Quick Updates

- I ran the analysis with the Thai data and got positive results for both the single zone and multi zone case.
- I added a proof that the multi-zone problem is a relaxation of the single zone problem.
- I talked to XY, Rad, and a potential 4th committee member.

Project Overview

- The goal of this project is to improve the design of index insurance contracts. I am particularly interested in the developing country setting.
- The original motivation was to develop a method to simultaneously design contracts for all insured zones, in order to better manage risk. However, upon learning more about the context I realized that the optimization based approach could be an improvement even in the single zone case.
- We develop a method that tries to maximize farmer utility, incorporates different kinds of constraints, and yields interpretable contracts.

Motivation for multi-zone optimization

- For the insurer, the risk (and therefor cost) of providing a line of insurance depends on the other lines of insurance in its portfolio.
- If all of its lines of insurance are highly correlated, it will be more expensive due to capital requirements.
- This is similar to the problem of portfolio design, where the correlations between stocks affect optimal portfolio composition.
- Insurers can influence the composition of their portfolio through the contracts they offer, so they should be better able to manage their costs if they take into account the correlation of different lines of insurance when designing contracts.

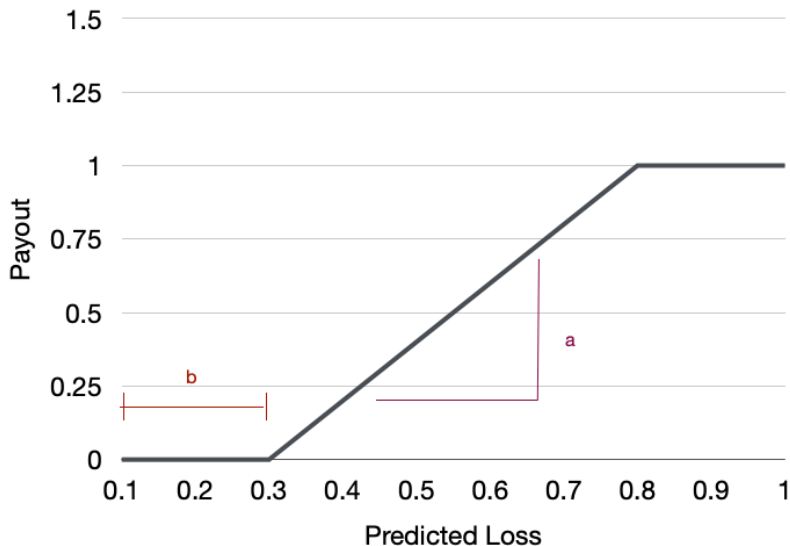
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Index Insurance: Definition and Parameters

- Index insurance uses a signal, θ , to predict agricultural loss, $\hat{\ell}(\theta)$
- Contract form: $I(\hat{\ell}(\theta)) = \min \left\{ \max \left\{ a\hat{\ell}(\theta) - b, 0 \right\}, 1 \right\}$, a, b are the contract parameters.

Example of Index Insurance Contract



Current methods to design index insurance

- **Baseline Method:** Predict then design, design contracts to maximize correlation between payouts and losses. Chantarat et al. (2013)
 - ▶ **Pros:** simple, interpretable
 - ▶ **Cons:** can't incorporate constraints
- **NN Based Method:** End to end, use NN to design contracts. Combine prediction and contract design into single step. Chen et al. (2023)
 - ▶ **Pros:** directly maximizes utility, admits price constraints
 - ▶ **Cons:** not interpretable, hard to adjust, requires a lot of data to train.

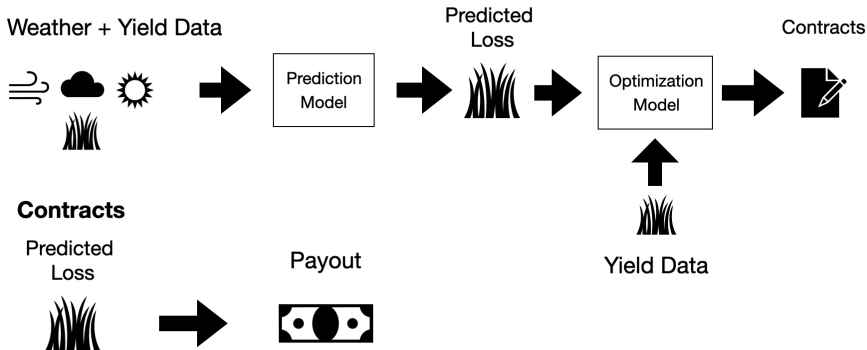
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- We opt for a "predict-then-optimize" approach.
- We use specialized time-series feature extraction algorithms and traditional ML algorithms (e.g. Random Forest, Gradient Boosting, Support Vector Machines) to build a loss prediction model.
- We then use model predictions and actual losses to design contracts to maximize utility.

Our Method Flowchart

Flowchart of our Method



Benefits of Our Approach

- Can admit many common definitions of the premium
- Can incorporate price and payout frequency constraints
- Requires less data to train than NN approach.

$$w = w_0 + 1 - \ell + I(\hat{\ell}) - \pi$$

$$\pi = \mathbb{E}[I(\hat{\ell})] + c_{\kappa} K$$

$$K = \text{CVaR}_{1-\epsilon} \left(I(\hat{\ell}) \right) - \mathbb{E}[I(\hat{\ell})]$$

CRRA Utility function: $u(w) = \frac{1}{1-\alpha} w^{1-\alpha}$. Premium definition comes from *Risk Modeling for Appraising Named Peril Index Insurance Products: A Guide for Practitioners* (Mapfumo, Groenendaal, and Dugger (2017)). The EU's Solvency requirements are similar.

Idealized Model

$$\max_{a,b,\pi,K} \mathbb{E} [U(w_0 + 1 - \ell - \pi + I(\theta))] \quad (1)$$

$$\text{s.t.} \quad I(\theta) = \min \left\{ \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}, 1 \right\} \quad (2)$$

$$\pi = \mathbb{E} [I(\theta)] + c_\kappa [\text{CVaR}_{1-\epsilon_K} (I(\theta)) - \mathbb{E}[I(\theta)]] \quad (3)$$

$$\pi \leq \bar{\pi}. \quad (4)$$

Convex Relaxation

$$\max_{a,b,K,\pi} \mathbb{E} \left[U \left(w_0 + 1 - \ell - \pi + \underline{I(\theta)} \right) \right] \quad (5)$$

$$\text{s.t.} \quad \overline{I(\theta)} = \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}$$

$$\underline{I(\theta)} = \min \left\{ a\hat{\ell}(\theta) + b, 1 \right\}$$

$$\pi = (1 - c_\kappa) \mathbb{E} \left[\overline{I(\theta)} \right] + c_\kappa \text{CVaR}_{1-\epsilon_K} \left(\overline{I(\theta)} \right) \quad (6)$$

$$\pi \leq \overline{\pi}.$$

Multiple Zone Case

In the multi-zone case, the problems are coupled through the required capital

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right]$$

The first term will be affected by the correlation of losses across zones, so the optimal contracts will vary depending on this correlation. There are also additional decision variables k_z .

Multiple Zone Model

$$\begin{aligned} \max_{a,b,K,\pi} \quad & \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z^j - \pi_z + I_z(\theta_z^j)) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[\overline{I_z(\theta_z)} \right] + \frac{c_K}{s_z} k_z \end{aligned} \quad (7)$$

$$K = \text{CVaR}_{1-\epsilon_K} \left(\sum_z s_z \overline{I_z(\theta_z)} \right) - \mathbb{E} \left[\sum_z s_z \underline{I_z(\theta_{z'})} \right] \quad (8)$$

$$K = \sum_z k_z \quad (9)$$

$$\pi_z \leq \overline{\pi_z} \quad (10)$$

$$\overline{I_z(\theta_z)} = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) + b_z \right\}$$

$$\underline{I_z(\theta_z)} = \min \left\{ a_z \hat{\ell}_z(\theta_z) + b_z, 1 \right\}$$

Single-zone vs Multi-zone

- $P1(z)$ - optimization program that designs contract for zone z independently of other zones.
- $P2$ - multi-zone optimization program.
- Let $P1(z)^*$ be the optimal value of $P1$ on zone z and let $P2^*$ be the optimal value of $P2$. We want to compare $\sum_z P1(z)^*$ and $P2$.
- We will show that $P2$ is a relaxation of running $P1$ for all zones and taking the sum. I.e. $P2^* \geq \sum_z P1(z)^*$.

SZ vs MZ: Constructing a solution

- Let $\mathbf{a}^{SZ} = (a_1^{SZ}, \dots, a_Z^{SZ})$ and $\mathbf{b}^{SZ} = (b_1^{SZ}, \dots, b_Z^{SZ})$ be the optimal values of a, b obtained from running $P1(z)$ for every zone, and let $\boldsymbol{\pi}^{SZ} = (\pi_1^{SZ}, \dots, \pi_Z^{SZ})$ and $\mathbf{K}^{SZ} = (K_1^{SZ}, \dots, K_Z^{SZ})$ be the corresponding values of π and K implied by a and b in each case.
- We denote our proposed solution for $P2$ with the superscript MZ .
- We set $\mathbf{a}^{MZ} = \mathbf{a}^{SZ}$ and $\mathbf{b}^{MZ} = \mathbf{b}^{SZ}$ and $\overline{K^{SZ}} = \sum_z s_z K_z^{SZ}$.
- Let K^{MZ} be the value of K in $P2$ implied by our proposed $\mathbf{a}^{MZ}$ and $\mathbf{b}^{MZ}$. Define $\Delta K = \overline{K^{SZ}} - K^{MZ}$. Set $k_z^{MZ} = s_z K_z^{SZ} - \frac{\Delta K}{|Z|}$.

SZ vs MZ: Feasibility

Constraints (7) and (8) hold automatically. Want to check $\sum_z k_z = K^{MZ}$.

$$\begin{aligned}\sum_z k_z^{MZ} &= \sum_z \left(s_z K_z^{SZ} - \frac{\Delta K}{|Z|} \right) \\ &= \sum_z s_z K_z^{SZ} - (\overline{K^{SZ}} - K^{MZ}) \\ &= K^{MZ}\end{aligned}$$

SZ vs MZ: Feasibility

Need to verify $\pi_z \leq \overline{\pi_z}$ for all z .

$$\begin{aligned}\pi_z^{SZ} &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + c_\kappa K_z^{SZ} \\ &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_k}{s_z} s_z K_z^{SZ} \\ \pi_z^{MZ} &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_\kappa}{s_z} k_z^{MZ} \\ &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_\kappa}{s_z} (s_z K_z^{SZ} - \frac{\Delta K}{|Z|})\end{aligned}$$

Constraint will be satisfied if $\Delta K \geq 0$

$$\Delta K \geq 0$$

$$\begin{aligned}\Delta K &= \overline{K^{SZ}} - K^{MZ} \\ \overline{K^{SZ}} &= \sum_z s_z K_z^{SZ} \\ &= \sum_z s_z \left[\text{CVaR}(I_z(\hat{\ell}_z)) - \mathbb{E} \left[I_z(\hat{\ell}_z) \right] \right] \\ &= \sum_z \text{CVaR}(s_z I_z(\hat{\ell}_z)) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right] \\ &\geq \text{CVaR} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right] \\ &= K^{MZ}\end{aligned}$$

SZ vs MZ Conclusion

$\ell_z, \hat{\ell}_z, I_z(\hat{\ell}_z)$ and w_0 are the same in both objectives. The only difference between the objectives of $P2$ and $P1$ is π^{MZ} and π^{SZ} . Since $u()$ is monotonically increasing, we have that $P2 \geq \sum_z P1(z)$

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Evaluation: Thai Data

- Index insurance is especially popular in developing countries because of its low cost.
- However, one of the difficulties in studying index insurance is the lack of data in these settings.
- I collaborated with the Bank of Thailand and was given access to a novel dataset of census-tract-level (Tambon) losses from natural disasters.

Thai Data: Administrative Data Sources

- Thailand Department of Agricultural Extension (DOAE): This dataset contains detailed information on the planting activities of over 3.1 million Thai farmers. Information includes plant date, plot size, rice variety, etc
- Thai Government Disaster Relief Data: This data is from the current national disaster relief program. It contains information about the disaster type, losses, and disaster date.

Remote Sensing Data Sources

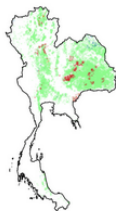
- Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS): daily rainfall data 1981-present
- MOD21A1 Land Surface Temperature: daily temperature data, available 2000-present
- FLDAS Famine Early Warning Systems Network: monthly evapotranspiration data, available 1992-present

Data Summary Statistics

Characteristic	Overall N = 49,052 [†]	2015 N = 6,037 [†]	2016 N = 6,358 [†]	2017 N = 6,285 [†]	2018 N = 6,288 [†]	2019 N = 6,268 [†]	2020 N = 6,079 [†]	2021 N = 5,510 [†]	2022 N = 6,227 [†]
Loss	0.02 (0.08)	0.00 (0.03)	0.02 (0.08)	0.04 (0.11)	0.01 (0.05)	0.01 (0.07)	0.00 (0.03)	0.03 (0.10)	0.03 (0.11)
Zone									
C1	6,052 (12%)	743 (12%)	763 (12%)	757 (12%)	761 (12%)	762 (12%)	766 (13%)	737 (13%)	763 (12%)
C2	3,064 (6.2%)	366 (6.1%)	404 (6.4%)	394 (6.3%)	393 (6.3%)	396 (6.3%)	391 (6.4%)	329 (6.0%)	391 (6.3%)
C3	3,139 (6.4%)	392 (6.5%)	426 (6.7%)	405 (6.4%)	410 (6.5%)	410 (6.5%)	398 (6.5%)	294 (5.3%)	404 (6.5%)
N1	6,312 (13%)	784 (13%)	808 (13%)	805 (13%)	803 (13%)	798 (13%)	798 (13%)	718 (13%)	798 (13%)
N2	3,593 (7.3%)	451 (7.5%)	451 (7.1%)	454 (7.2%)	449 (7.1%)	451 (7.2%)	450 (7.4%)	437 (7.9%)	450 (7.2%)
N3	2,105 (4.3%)	269 (4.5%)	268 (4.2%)	268 (4.3%)	265 (4.2%)	264 (4.2%)	265 (4.4%)	241 (4.4%)	265 (4.3%)
NE1	1,294 (2.6%)	162 (2.7%)	162 (2.5%)	162 (2.6%)	162 (2.6%)	162 (2.6%)	162 (2.7%)	160 (2.9%)	162 (2.6%)
NE2	9,400 (19%)	1,178 (20%)	1,181 (19%)	1,183 (19%)	1,183 (19%)	1,179 (19%)	1,178 (19%)	1,138 (21%)	1,180 (19%)
NE3	10,546 (21%)	1,317 (22%)	1,320 (21%)	1,319 (21%)	1,319 (21%)	1,320 (21%)	1,319 (22%)	1,315 (24%)	1,317 (21%)
S1	1,025 (2.1%)	111 (1.8%)	145 (2.3%)	141 (2.2%)	140 (2.2%)	148 (2.4%)	127 (2.1%)	76 (1.4%)	137 (2.2%)
S2	2,522 (5.1%)	264 (4.4%)	430 (6.8%)	397 (6.3%)	403 (6.4%)	378 (6.0%)	225 (3.7%)	65 (1.2%)	360 (5.8%)

[†] Mean (SD); n (%)

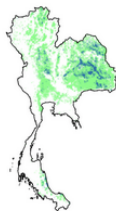
Disaster Activity Map



2015



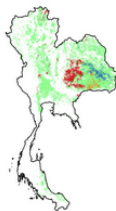
2016



2017



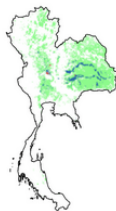
2018



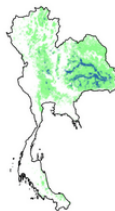
2019



2020



2021



2022

Evaluation Procedure

- We use leave one year out cross validation.
- In each CV fold, data from year y is used for testing. Data from $y' \neq y$ is used for insurance design (prediction model training and contract design).
- This gives us out of sample predictions for every observation in our data. We then compute utility-based performance metrics across all out of sample predictions.

Evaluation Details

- We use training data to compute premium $\rightarrow$ this reflects practice.
- We modify Chen's method to use the same premium definition as ours.
- We use a copula approach to estimate the CVaR when calculating the premium.
- Same method is used to calculate the premium for all methods.
- We use a CRRA utility function with $\alpha = 1.5$.

Performance Metrics

- We focus on the Relative Insurance Benefit (RIB) as our main metric.
- Used in index insurance studies (Kenduiywo et al. (2021); Estefania-Salazar et al. (2024))
- Compares a proposed insurance contract to perfect insurance. It measures the welfare gains from perfect insurance that the proposed contract recovers.
- Interpretation: if RIB of an insurance J is 50%, we can think of it as meaning that it is 50% as good as perfect insurance. In other words, it yields half as many welfare gains as perfect insurance.

RIB Definition

Let w_i^J be the wealth of farmer i under insurance contract J . We denote perfect insurance with PI and no insurance with NI

- $CE(J)$: $u^{-1}(\mathbb{E}[u(w^J)])$
- $\Delta CE(J) := CE(J) - CE(NI)$
- $RIB(J)$: $\frac{\Delta CE(J)}{\Delta CE(PI)}$
- Utility: $u(w) = \frac{1}{1-\alpha} w^{\frac{1}{1-\alpha}}$

Single Zone Results

Here, the contract design and pricing for each zone happens independently of all other zones. Premium pricing is also independent:

$$\pi_z = \mathbb{E}[I_z(\theta_z)] + c_k K_z, \text{ with } K_z = \text{CVaR}_{1-\epsilon_K}(I_z(\theta_z)) - \mathbb{E}[I_z(\theta_z)]$$

Method	RIB	Premium	Cost	Cost PI
VMX	0.679	0.014	0.017	0.027
Chantararat	-0.244	0.004	0.002	0.027
Chen	-2.676	0.073	0.062	0.027

Single Zone Results Key Points

- Our method achieves 68% of the utility gains of perfect insurance.
- Has a much lower premium and cost than Chen's method.

Note on the value of c_K

- Potential value for farmers is highly dependent on c_K and risk aversion, α
- For the most commonly used value of α , farmers could only benefit from perfect insurance if $c_K = 0.02$.
- As risk aversion increases, the gains from perfect insurance increase.

Effects of Cost of Capital and Risk Aversion

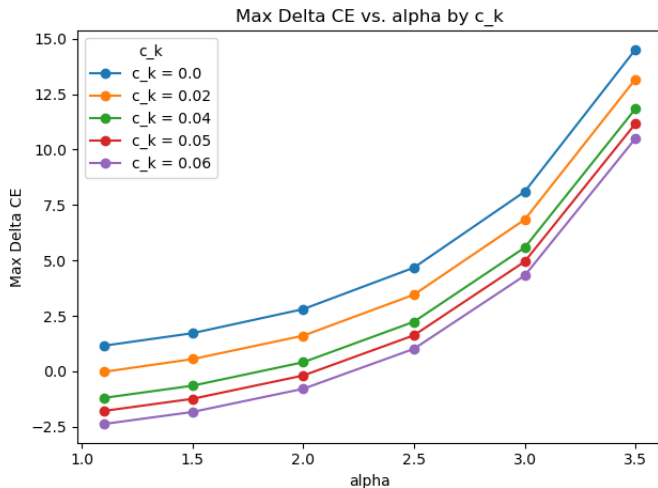


Figure: Perfect Insurance

In MZ case premium is based on portfolio required capital:

$$K = \text{CVaR}_{1-\epsilon_K} \left(\sum_z s_z \overline{I_z(\theta_z)} \right) - \mathbb{E} \left[\sum_z s_z \underline{I_z(\theta_{z'})} \right].$$

Multi zone results

Method	RIB	Premium	Cost	Cost PI
VMX-M	0.592	0.035	0.036	0.027
VMX	0.524	0.014	0.017	0.027
Chantararat	-0.163	0.004	0.002	0.027
Chen	-1.909	0.074	0.062	0.027

- Payout Precision: share of all payouts that are "valid".

$$P = \frac{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i \min\{\ell_i, \hat{y}_i\}}{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i}$$

- Loss Recall: ratio of valid payouts to loss. $R = \frac{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i \min\{\ell_i, \hat{y}_i\}}{\sum_{i:\hat{y}_i > 0} w_i \ell_i}$

Valid Gains and Bad Losses

	Utility Gain	Utility Loss
Loss	A	B
No Loss	C	D

A: utility gain during loss → "valid gain"

B: utility loss during loss → "bad loss"

Insurance Behavior Results

Method	RIB	Payout Precision	Loss Recall	Valid Gain	Bad Loss
VMX-M	0.592	0.184	0.211	0.390	0.334
VMX	0.524	0.207	0.128	0.379	0.431
Chantararat	-0.163	0.338	0.021	0.594	0.465
Chen	-1.909	0.173	0.298	0.329	0.440

Summary of results

- Our method leads to larger utility gains.
- Has more efficient behavior.
- Lower cost.

References

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Idealized CVaR Model

- **Objective:** conditional value at risk of the farmers' loss net of insurance.
- **Constraint 1:** piecewise linear structure of the contract.
- **Constraint 2:** budget constraint.
- **Constraint 3:** definition of required capital.

$$\min_{a,b,\pi,K} \text{CVaR}_{1-\epsilon}(\ell - I(\theta))$$

$$\text{s.t. } I(\theta) = \min\{(a\hat{\ell}(\theta) + b)^+, P\} \quad (11)$$

$$\mathbb{E}[I(\theta)] + c_\kappa K \leq B \quad (12)$$

$$K = (\text{CVaR}_{1-\epsilon}(I(\theta)) - \mathbb{E}[I(\theta)]) \quad (13)$$

The problem is non-convex, so we need convex approximations

We use the following approximations of $I(\theta)$ to make the problem convex:

$$\begin{aligned}\overline{I(\theta)} &\triangleq \max \left\{ 0, a\hat{\ell}(\theta) + b \right\} \\ \underline{I(\theta)} &\triangleq \min \{ a\hat{\ell}(\theta) + b, K \}\end{aligned}$$

- Note that $\overline{I(\theta)} \geq I(\theta)$ and $\overline{I(\theta)}$ is convex. Conversely, $\underline{I(\theta)} \leq I(\theta)$ and $\underline{I(\theta)}$ is concave.
- We replace $I(\theta)$ with either $\overline{I(\theta)}$ or $\underline{I(\theta)}$ where necessary to obtain conservative and convex approximations.
- We also need approximations or proxies for $E[I(\theta)]$ in constraint . We use $\pi_{SQ} = E[I_{SQ}(\theta)]$, where I_{SQ} is the contract designed using the status quo method, as a proxy for $E[I(\theta)]$ in constraint .